

Birla Institute of Technology & Science, Pilani, K. K. BIRLA Goa
campus Database Systems (CS F212) Second Semester 2023-2024
Lab No. 6 Topic: Joins and Cross Product

Time: 3 TO 4:50 PM

Instructions about file name conventions:

Download the IDNO_Lab6.sql file. Rename it with your id number. Copy all the SQL queries in this file.

The file name must be: <ID.NO.>_Lab<LabNo.>.sql>

Example: 2022A7PS0011G_Lab6.sql

Instruction about the format of the answers:

Each SQL query must end with a semicolon.

Do not include the Question Number, any comments, or any NonSQL text in your .sql file. Type ; for the question you don't want to answer.

Instructions to upload:

Upload the .sql file that has all your SQL queries

The upload link will be activated at 4:50 for 3 minutes.

NOTE:

Do not Set operations in today's lab.

Incorrect File names, incorrect file extensions, and incorrect file formats will not be evaluated.

Import the following.sql files to solve the following questions.

Lab6tables.sql

Exercise

Write SQL queries to display the following information. For each question write a **single SQL** query.

Q1. Find all possible ordered pairs (s1,s2) where s1 and s2 are students and s1 and s2 belong to the same batch(year). **[1M]**

Expected output:

```
+-----+-----+
| student1_name | student2_name |
+-----+-----+
| Gautam Jhabak | Shashank Rana |
```

Shashank Rana Gautam Jhabak
Rhythm Chauhan Sakshi Singh
Nandish Choksi Prateek Chitre
Himanshu Singh Prateek Chitre
Aryan Nambiar Prateek Chitre
Nandish Choksi Aryan Nambiar
Himanshu Singh Aryan Nambiar
Prateek Chitre Aryan Nambiar
Sakshi Singh Rhythm Chauhan
Nandish Choksi Himanshu Singh
Aryan Nambiar Himanshu Singh
Prateek Chitre Himanshu Singh
Himanshu Singh Nandish Choksi
Aryan Nambiar Nandish Choksi
Prateek Chitre Nandish Choksi
+-----+-----+

Q2. Retrieve the names of all students along with the names of their assigned teachers.
[1M] Expected output:

+-----+-----+
student_name teacher_name
+-----+-----+
Shashank Rana Love Babbar
Sakshi Singh Aditya Verma
Prateek Chitre Alakh Pandey
Prateek Chitre Gate Smashers
Ujjwal Kumar Code with Harry
Ujjwal Kumar Shraddha Didi
Aryan Nambiar Neso Academy
Aryan Nambiar Shraddha Didi
Ayush Ghatalia Code with Harry
Himanshu Singh Gate Smashers
+-----+-----+

Q3. List all students and their grades(if any), along with their teacher's name(if any) in descending order of grades.' Please refer to the expected outputs. if the student does not have a grade/teacher assigned, include the student name but assign NULL in the grade and teacher_name column. **[1M]**
Expected output:

```

+-----+-----+-----+
| student_name | grade | teacher_name |
+-----+-----+-----+
| Prateek Chitre | 95.25 | Gate Smashers |
| Aryan Nambiar | 92.75 | Neso Academy |
| Sakshi Singh | 90.00 | Aditya Verma |
| Ayush Ghatalia | 88.75 | Code with Harry |
| Shashank Rana | 87.00 | Love Babbar |
| Himanshu Singh | 86.50 | Gate Smashers |
| Prateek Chitre | 85.50 | Alakh Pandey |
| Ujjwal Kumar | 82.40 | Shraddha Didi |
| Ujjwal Kumar | 80.90 | Code with Harry |
| Aryan Nambiar | 78.30 | Shraddha Didi |
| Gautam Jhabak | NULL | NULL |
| Rhythm Chauhan | NULL | NULL |
| Nandish Choksi | NULL | NULL |
+-----+-----+-----+

```

Q4. List the average grade of each student along with the name of the student. List only those students who have an average grade greater than 85. **[2M]**

Expected output:

```

+-----+-----+-----+
| student_id | student_name | avg_grade |
+-----+-----+-----+
| S2345 | Prateek Chitre | 90.375000 |
| S3456 | Aryan Nambiar | 85.525000 |
| S1456 | Shashank Rana | 87.000000 |
| S3789 | Ayush Ghatalia | 88.750000 |
| S2101 | Sakshi Singh | 90.000000 |
| S4123 | Himanshu Singh | 86.500000 |
+-----+-----+-----+

```

Q5. Find the teachers (teacher id and teacher name) who neither have any ongoing assignments, nor any ongoing projects. **[1M]**

Expected output:

```

+---+-----+
| id | name |
+---+-----+
| 10 | Sandeep Jain |
+---+-----+

```

Q6. Find the youngest teacher (id, name, age) who is teaching mathematics.

[1M] Expected output:

```
+----+-----+----+
| id | name | age |
+----+-----+----+
| 4  | Namo Kaul | 28 |
+----+-----+----+
```

Q7. Find the name of the teacher that requires the maximum project budget in the Mathematics course. **[1M]**

Expected output:

```
+-----+
| name |
+-----+
| Alakh Pandey |
+-----+
```

Q8. Among all the students who have submitted their assignments in the month of March, find the student (student id and student name) with the lowest grade. Solve without using group by, order by, limit clause, and aggregate functions. **[2M]**

Expected output:

```
+-----+-----+
| id | name |
+-----+-----+
| S2777 | Ujjwal Kumar |
+-----+-----+
```